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A low-carbon transportation network: Collaborative effects of a rail freight subsidy and carbon trading mechanism

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ABSTRACT

Optimizing a transportation network is an effective way to reduce carbon emissions. To examine the collaborative effects of a rail freight subsidy and carbon trading mechanism in a low-carbon transportation network, a multiobjective 0–1 mathematical model that considers transportation cost, carbon trading cost and transportation time is established in this paper, and the NSGA-II algorithm is used to solve it. The Pareto optimal frontier solution set is found for the model, and the optimal solution is determined using the evaluation function of the ideal point method. The performance and effectiveness of the NSGA-II algorithm is analyzed by means of a sample example. A case study of the Yangtze River Economic Belt region in China is conducted to demonstrate the application and practicality of the model. Sensitivity analysis is carried out on rail freight subsidy, carbon quota and carbon trading price. The numerical results highlight that the rail freight subsidy significantly contributes to the design of the low-carbon transportation network, while the low carbon trading price shows a limited effect, which also leads to a weak effect of carbon quota on low-carbon transportation network design. These findings provide decision-making support for optimizing the low-carbon transportation network design and improving the carbon trading mechanism.

1. Introduction

To overcome resource and environmental constraints and promote high-quality and sustainable development, the Chinese government in 2020 proposed that the country reach its carbon peak by 2030 and be carbon neutral by 2060 (He et al., 2022; Shao et al., 2022; Zhang et al., 2023b). The transportation industry, as the third largest source of carbon dioxide (CO_2) emissions (Du et al., 2019), is a key sector for achieving these goals. Transportation constitutes 16.5 % of global greenhouse gas (GHG) emissions (Huang et al., 2023a). In terms of mode, road transportation is the industry's main source of carbon emissions (Liu et al., 2022). In 2019, road, rail, water and air transportation accounted for 84.7 %, 1.4 %, 6.9 % and 6.9 % of the total CO_2 emissions from transportation, respectively (ICCSDTU et al., 2022). The heavy reliance on road transportation and the inefficient structure of the transportation network greatly affect carbon emissions in the industry. Therefore, adopting intermodal transportation is an effective way to save energy and reduce

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emissions as the average carbon intensity of intermodal transportation is lower than that of trucking by 46 % (Craig et al., 2013). Such restructuring of the transportation mode and network can help China meet its dual carbon targets. The principles of reduction in carbon emissions via intermodal transportation are shown in Fig. 1.

To effectively control total GHG emissions and fight climate change, various regulations and market-based mechanisms have been introduced in the past decades which have had significant impacts on the transportation sector (Zheng et al., 2017; Zheng et al., 2021; Zheng et al., 2022; Zheng et al., 2023; Tan et al., 2023). The European Union introduced the emissions trading system (ETS) in 2005 (Toptal et al., 2014; Ferrara and Giua, 2022). In 2011, the Chinese government launched a similar carbon trading system to encourage companies to reduce their carbon emissions (Chen et al., 2022). The carbon quota and trading mechanisms transform unconstrained emissions into limited resources, which motivate polluting enterprises to reduce their consumption and emissions. Thus, an ETS is an important tool for achieving China's dual carbon targets (Jiang, 2014). However, China currently includes only fixed emission sources, such as ports and airports, and public mobile emission sources in its carbon trading system, involving cities such as Shanghai, Shenzhen and Beijing (Li and Tang, 2017). The economic pressure of an ETS also forces transport enterprises to adjust their operations and promotes structural transformation and technological reengineering. However, during the low-carbon transformation process, transport enterprises suffer from those problems such as low adaptability and insufficient competitiveness. Currently, China's ETS imposes a great burden on enterprises, preventing them from obtaining reasonable profits and adequate revenue, which affects the quality of services. Additionally, the inclusion of the transportation industry in the carbon trading mechanism is still at an early stage, with problems such as incomprehensive coverage and inconsistent allocation of carbon quotas. These problems restrict the ability of the carbon trading system to reduce emissions and weaken the motivation of high-emission enterprises to engage in low-carbon transformation.

To accelerate the low-carbon and sustainable transformation of the transportation industry, subsidizing freight transportation for low-carbon modes is an effective way to rationalize the transportation mode choice and network configuration (Chen et al., 2023). In an unbalanced transportation network, the development of alternative modalities can enhance competitiveness (Larranaga et al., 2017) and thus help reduce cost and emissions. Meanwhile, freight subsidies can help alleviate economic conflict, support the transportation industry during its low-carbon transformation and motivate enterprises to make operational improvements and technological innovations achieve supply-side structural reform. Thus, it is an effective way to promote the sustainable development of the freight transportation system.

Some European countries and the United States have adopted subsidy policies to fully exploit the potential of intermodal transportation. The Belgian government's annual budget includes €30 million for subsidies to intermodal operators to maintain rail transportation levels, including a subsidy scheme for container transportation at inland waterway terminals in the Brussels Capital Region (Macharis and Pekin, 2009). The German government enacted an intermodal transportation subsidy to support the shift of freight transportation from roads to more environmentally friendly transportation modes, such as rail and waterway. The United States government has adopted subsidy measures to promote coastal shipping (USDOT, 2011). Domestically, the Chinese government has adopted a series of rail freight subsidies for China Railway Express (CRE) to bridge the price gap between CRE and China–EU maritime

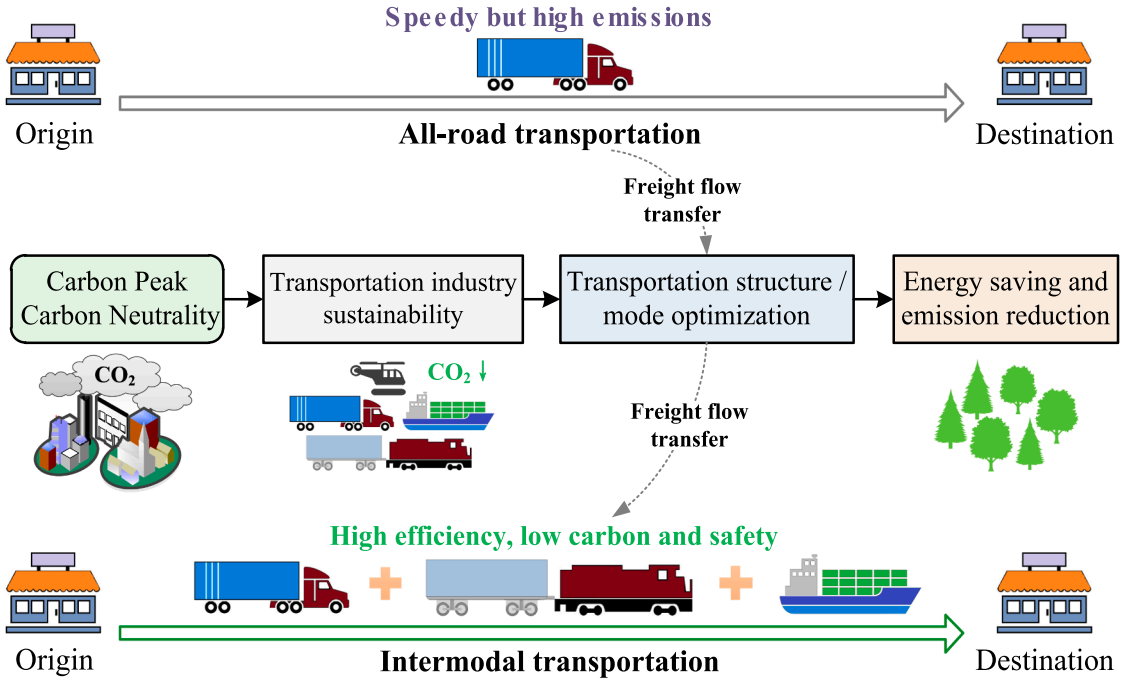


Fig. 1. Principles of reduction in carbon emissions via intermodal transportation.

transportation (Jiang et al., 2018; Yin et al., 2020) and enhance the CRE's market competitiveness. In addition, since January 1, 2018, rail enterprises have been able to independently determine their freight levels in accordance with market supply and demand within a range of 15 % rise and unlimited fall according to the benchmark freight rate set by the Chinese government. Compared with road transportation, the proportion of freight transported by rail is relatively low. The introduction of a rail freight subsidy could promote the development of rail transportation and further rationalize the freight transportation structure. Thus, the problem of high carbon emissions from transportation caused by the high proportion of road transportation can also be solved. Although waterway transportation should be subsidized because of its low-carbon and environmental advantages, China's main inland waterway for transportation, the Yangtze River, has a serious congestion problem (Yan et al., 2017). A subsidy to waterway transportation would aggravate congestion and further lead to problems such as poor timeliness of freight transportation. In addition, the insufficient carrying capacity of the Three Gorges Dam on the Yangtze River has led to long waiting time for ships, and the bottleneck problem of inland waterway transportation is even more prominent (Yin et al., 2023). Rail transportation also has a significant low-carbon and sustainable advantage (Sina Mohri and Thompson, 2022) that aligns with the industry's energy-saving and emission-reduction goals. More importantly, with the development of high-speed railways in China, passenger and cargo transportation patterns have gradually diverged, releasing capacity for freight transportation. Therefore, with the limited emissions reduction effect of the carbon trading mechanism, introducing a rail freight subsidy for intermodal transportation could further promote healthy competition and sustainable development of the freight market and the efficient allocation of low-carbon transportation resources.

Based on field investigation, the current freight subsidy policies introduced by the Chinese government on intermodal transportation tend to lack sustainability (Xu et al., 2018). When the subsidies are eliminated, the share of intermodal transportation may decline, further affects the carbon reduction of transportation industry. Meanwhile, China is currently setting a dual carbon target, while the carbon trading mechanism is still immature (Lin and Jia, 2018; Zhang and Zhang, 2021), and the current low carbon trading price is not sufficient to influence decisions on transportation schemes (Zhong et al., 2023; Wu et al., 2023). To address these concerns, this study analyzes the transportation network optimization problem by simultaneously considering the effects of a freight subsidy and the carbon trading mechanism. This approach conforms with China's policy orientation for the low-carbon transformation of its transportation industry, and therefore contributes to the development of a comprehensive high-quality transportation system. The contribution of this paper includes:

(1) The development of a freight subsidy in conjunction with the carbon trading mechanism in accordance with China's low-carbon sustainable development goals. The study aims to address the realities of weak sustainability of the freight subsidy policy and immaturity of the carbon trading mechanism, and thus, offers strong practical value.
(2) The transportation network design problem under the joint influence of a rail freight subsidy and the carbon trading mechanism is analyzed, and the optimal transportation scheme including transportation paths and modes is identified using the evaluation function of the ideal point method to weigh transportation cost, carbon trading cost and transportation time. Technically, a multiobjective 0–1 mathematical model is constructed, and the fast elitist non-dominated sorting genetic algorithm (NSGA-II) is used to solve the model. Meanwhile, the performance of the solution method is verified with a numerical example.
(3) A real-world case study in the Yangtze River Economic Belt, China is presented to verify the application and practicality of the model and sensitivity analyses are conducted to further reflect the collaborative effects of a rail freight subsidy and carbon trading mechanism on intermodal transportation network design and achieve generalize results.
(4) The modelling work leads to the policy recommendations for different actors of transport service. Meanwhile, policy implications are gained for government departments : a freight subsidy mechanism for low-carbon transportation modes, adjustment of the freight market structure via the carbon trading price mechanism and a scientific formulation for the allocation of carbon quotas for high-emission enterprises in the transportation industry. Besides, a comprehensive analysis of the applicability and transferability of new findings is presented.

The remainder of this paper is organized as follows. Section 2 reviews the literature related to low-carbon transportation network optimization, carbon trading mechanism and freight subsidies in the transportation industry. Section 3 illustrates the multiobjective optimization model of an intermodal transportation network with a rail freight subsidy and carbon trading mechanism, and the NSGA-II solution algorithm is analyzed in Section 4. In Section 5, a small-scale numerical example is presented to analyze the performance of the solution method, and a real-world case study in China is conducted to demonstrate the application and practicality of the model. Section 6 offers policy implications for the design and development of low-carbon transportation based on the results. Section 7 summarizes the conclusions of this study and suggests future research directions.

2. Literature review

The literature on the optimization of low-carbon transportation networks mainly focuses on a carbon tax, trading mechanism or other methods to control and reduce emissions from freight transportation to promote the low-carbon development of the transportation industry. Some studies incorporate carbon emissions as a constraint (Wang et al., 2015; Lam and Gu, 2016; Sheng et al., 2017; Dai et al., 2018) or objective function (Yin et al., 2021b; Yang et al., 2021; Yin et al., 2022) into the model to reduce carbon emissions and optimize the transportation network. As a carbon reduction strategy, a carbon tax converts carbon emissions into carbon emission costs. Such a cost constraint introduces incentives to optimize the transportation network, save energy and reduce emissions. Unlike a carbon tax that taxes whoever emits, a carbon trading mechanism sets a limit on total carbon emissions, which adjusts the transportation structure and improves transportation management decisions via a market mechanism. Rosič and Jammernegg (2013)

Table 1
Summary of relevant studies.

Question category	Author & Year of Publication	Methodology	Contribution
Low-carbon transportation network optimization	Wang et al. (2015)	An economic model and benchmark case	Facilitates policy makers to balance the effects on stakeholder when designing ETS schemes
	Lam and Gu (2016)	A bi-objective optimization model	Practical insights into the trade-offs between cost and transit time and the impact of carbon emission limits
	Sheng et al. (2017)	A two-stage game	Unilateral regulation may lead to an increase in total emissions
	Dai et al. (2018)	An economic model and alternative shipping networks	Shows that if emission charge is above a threshold, carriers will reconfigure networks to minimize costs including emission charge
	Yin et al. (2021b)	A multiobjective 0–1 mathematical model with non-dominated ranking and ideal point methods	Decision support for government to optimize transportation structures
	Yang et al. (2021)	A programming model solved with a local optimization algorithm	Network design and toll policies can reduce CO_2 emissions
	Yin et al. (2022)	The ideal point method to transform dual objectives into the single objective model	Decision support for the collaborative development of intermodal transportation with regional integration
	Rosič and Jammernegg (2013)	The dual sourcing model based on the newsvendor framework	Decision support for individual companies and policy implications for environmental regulations
	Demir et al. (2016)	A sample average approximation method	Proposes a capacitated multi-commodity green service network model
Effect assessment of low-carbon transportation networks	Ortegon-Sanchez and Oviedo Hernandez (2016)	A multidimensional analysis	Provides methodological and empirical contributions for low-carbon travel capacity
	Xu et al. (2022)	Non-radial directional distance function and social network analysis	Policy implications for energy conservation and emission reduction in China's transportation industry cross region
	Kim and Van Wee (2009)	Semi-life cycle assessment technique	Rail intermodal freight systems emit less CO_2 than truck-only systems
	Van Fan et al. (2018)	Air emission assessments	Highlights the importance of including air pollutants rather than limiting the focus on GHG emissions
	Yu et al. (2023)	Bayesian neural networks and SHAP models	Provides references for improving air quality from transportation plan
	Zhang et al. (2023)	Cross efficiency DEA model	Intermodal transportation can reduce transportation carbon emissions
Carbon quota trading mechanism	Chang and Lai (2013)	Mathematical programming methods with a semi-life cycle assessment and the top down method	Confirms the carbon allowance allocation policy's ability to reduce carbon emissions in the transportation industry
	Han et al. (2017)	A gray forecast model, trend extrapolation and top-down approach	A carbon emission trading mechanism for the transportation sector
	Shu et al. (2017)	An inventory cost model with carbon constraints	Practical insights regarding the allocation of carbon emission rights
	Li et al. (2018)	A game theoretical model	Identifies the need for policies to improve consumers' low-carbon preferences to maximize social welfare
	Li et al. (2020)	A DEA model with baseline excitation allocation	Validates a model with carbon quotas for road transportation in China
	Zhang et al. (2021b)	A bi-level programming method solved by a cloud particle swarm optimization algorithm	Suggestions for the location of new distribution centers and ideas for energy conservation and emissions reduction
	Zhu et al. (2023)	A bi-level model solved by a multiobjective particle swarm optimization algorithm	Confirms the effectiveness of carbon emission allowances and provides implications for multiple stakeholders
Freight subsidy mechanism	Zhang et al. (2018)	A bi-level formulation model with a genetic and Frank-Wolfe hybrid algorithm	Joint selection of logistics infrastructure investments and green subsidies is more effective than either approach alone
	Kundu and Sheu (2019)	A competition model based on game theory	Describes shippers' mode switching patterns in reply to a mode subsidy
	Yang et al. (2019)	The minimum cost flow model and a bi-level programming model	Shows that a coordinated subsidy achieves better results and better total social welfare than an uncoordinated one
	Li and Zhang (2020)	Lyapunov optimization to reduce the complexity of a multi-slot global optimization problem	Shows that sustainable rail transportation development requires improving service levels
	Lu et al. (2021)	A network optimization model and an adaptive genetic-simulated annealing algorithm	Shows that increasing train speed is a better way to improve the competitiveness of rail cold chain logistics than a freight subsidy
	Hu et al. (2022)	A bi-level programming model	Compares subsidy schemes, identifies the optimal scheme's environmental impacts

studied a dual sourcing model with emission trading and emission tax policies and found that from the management perspective, an emission trading mechanism is preferable to an emission tax. Demir et al. (2016) analyzed a green intermodal service network design problem considering transportation time uncertainty and the stochastic approach can develop robust transportation plans based on cost, time, and GHG emissions objectives. In addition, many studies have focused on effect assessment of low-carbon transportation networks, which also contribute to the further optimization of the transportation structure and sustainable development in transportation industry (Ortegon-Sanchez and Oviedo Hernandez, 2016; Xu et al., 2022). Kim and Van Wee (2009) assessed emissions from freight transport systems and found that rail-based intermodal freight systems emit less CO_2 than truck-only systems. Van Fan et al. (2018) emphasized the necessity of considering air pollutants for decision making and analyzed the air emissions assessment in transportation. Yu et al. (2023) evaluated the impact of road network topology on the reduction effect of $PM_{2.5}$ concentration, and provided reference for improving air quality from the perspective of transportation planning. Zhang et al. (2023) proposed a novel cross efficiency data envelopment analysis (DEA) model and evaluated the low carbon efficiency of rail-water transportation to ensure the sustainable development. These studies show that carbon emission restrictions, a carbon tax and a carbon trading mechanism all promote the low-carbon transformation of the transportation industry. They effectively reduce carbon emissions and save energy.

The carbon quota (allowance) mechanism is an effective policy instrument to mitigate carbon emissions in transportation (Chang and Lai, 2013). Han et al. (2017) analyzed carbon quotas in China's road transportation sector and designed a carbon emissions trading mechanism. Shu et al. (2017) constructed a manufacturer's inventory and production model based on a cap-and-trade regulation to analyze the influence of factors such as carbon quotas and trading prices on manufacturers' production decisions. Li et al. (2018) used a game theory model to analyze the changes in manufacturers' operational decisions under the cap-and-trade mechanisms, and suggested that strict pollution control policies can ensure that companies sufficiently reduce their emissions. From the carbon intensity perspective, Li et al. (2020) used DEA to develop a carbon emissions quota model for road transportation and applied the model to Chinese provinces and cities. Zhang et al. (2021b) constructed a model for the location of cold chain logistics facilities to analyze the impact of a carbon quota on total carbon emissions, total social costs and the scale of cold chain distribution centers. Zhu et al. (2023) presented the carbon allowance allocation in maritime shipping with a bi-level model. Recent studies emphasize the role of carbon quotas and trading mechanisms in the low-carbon transformation of enterprises' operations, providing a reference for the application of a carbon quota in the low-carbon development of transport enterprises.

To examine the effects of a freight subsidy, Zhang et al. (2018) constructed a bi-level programming model to optimize logistics infrastructure investment and green transportation mode subsidies and reduce carbon emissions in freight transportation. Kundu and Sheu (2019) constructed a competition model between rail and maritime transportation to analyze the impact of government subsidies on shippers' mode choices and proposed a subsidy scheme. Yang et al. (2019) constructed a bi-level programming model to identify the optimal subsidy in the context of uncoordinated subsidies for rail containers. Li and Zhang (2020) analyzed the effects of improving rail freight service quality and low-carbon subsidies on transportation mode transformation and proposed a framework to reduce CO_2 emissions from freight transportation activities. Lu et al. (2021) established an optimization model for the rail cold chain logistics network with freight subsidies and studied the effect of market share improvements through a sensitivity analysis of factors such as a rail freight subsidy and train speed. Hu et al. (2022) considered how to maximize intermodal transportation and minimize subsidy expenditures and constructed a bi-level programming model to develop the optimal subsidy scheme and maximize waterway transportation in an intermodal system. These studies show that a freight subsidy mechanism can enhance the market competitiveness of low-carbon transport modes and promote a modal shift and the collaborative development of transport modes.

Studies on the optimization of low-carbon transportation networks, carbon trading policies and freight subsidies are well developed. The relevant studies are summarized in Table 1. However, these studies have not fully addressed the following problems:

(1) Currently, the two main carbon pricing mechanisms are carbon trading and a carbon tax. The transportation sector of China is under a carbon trading mechanism but not a widely adopted carbon tax yet, which can create problems because of its inflexibility. In some studies, the quantification of carbon emission cost does not consider regional differences in carbon pricing policies. Instead, a carbon tax rate is used to convert carbon emissions into carbon emission costs, reducing the practical usefulness of the findings from these studies. In addition, some studies ignore carbon emissions during transshipment, resulting in a gap between the measured and actual carbon emissions.
(2) The rail freight subsidy mechanism is still immature, and most of the relevant studies focus on qualitative research. A scientific method to calculate the optimal freight subsidy rates for intermodal transportation is lacking, and the timing of the introduction of a freight subsidy mechanism and its effect on the transportation network structure requires further research.
(3) The introduction of freight subsidies for low-carbon transportation modes and a rational carbon quota mechanism can both contribute to controlling and reducing carbon emissions in the transportation industry. However, most studies focus on one aspect, so there is a lack of impactful research on the collaborative/joint effects of a freight subsidy and carbon trading mechanism.

In contrast, this paper considers China's actual carbon trading mechanism, introduces a mechanism to convert carbon emission trades into costs and establishes a multiobjective 0–1 mathematical model of the intermodal transportation network with coordinated rail freight subsidies and a carbon trading mechanism solved by the NSGA-II algorithm. A numerical sample example is used to analyze the performance and effectiveness of the NSGA-II algorithm, and the effects of various rail freight subsidy rates, carbon quota quantities and carbon trading prices on the intermodal transportation network are compared by a real-world case in China and the corresponding transportation network optimization scheme is obtained to improve the efficiency of intermodal transportation and reduce its carbon emissions. Finally, policy implications are proposed in terms of the design of efficient and low-carbon transportation

networks via the coordination of a freight subsidy and carbon trading mechanism. Such an analysis extends previous studies to investigate a potentially more effective and more comprehensive policy package.

3. Modelling methodology

Fig. 2 illustrates the combined influence of a rail freight subsidy and carbon trading mechanism on an intermodal transportation network. A transport enterprise has cargo to be transported from *O* to *D*. During transportation, it passes through several cities (nodes), and different transportation modes are available between adjacent nodes. The unit transportation cost, transportation time and carbon emissions of the modes differ, and if transshipment is carried out, the corresponding transshipment cost, carbon emissions and time are considered. If the enterprise chooses rail transportation, it receives the corresponding subsidy. Meanwhile, under the carbon trading mechanism, each enterprise has a certain carbon emissions quota. If an enterprise exceeds its quota, it must buy additional carbon emission permits through the carbon trading market, which generates carbon trading cost. The cost of purchasing carbon emissions quota may not be the same as the cost of selling carbon emissions quota in different regions, a feature that may be explicitly modelled if one focuses on the market mechanism of permit trading. However, the current study focuses on the cost of purchasing carbon emissions quota for transportation enterprises. Therefore, neither the sale of excess carbon emissions quota nor the specific mechanism used for carbon emission permit trading is modeled.

3.1. Assumptions

(1) The freight flow is indivisible during transportation and the transport mode can only be changed at a node.
(2) Speed and unit cost for each transport mode are the same between all pairs of nodes.
(3) Transshipment operations between the same mode of transport are not explicitly considered in the current specification (Yin et al., 2021; Guo et al., 2023; Zhang et al., 2024).

3.2. Notations

The detailed notations and descriptions are given in Table 2.

3.3. Mathematical model

3.3.1. Objective function

The objective function of the model includes total transportation cost *C*, carbon trading cost *E* and total transportation time *T*. Since the optimization of transportation cost, carbon trading cost and transportation time is extremely important for transport enterprises, to better illustrate the impact of each objective on the practical transportation scheme, to balance the trade-offs between multiple objectives, and to satisfy the concerns and needs of different stakeholders, including transportation companies, customers, and the government, etc., on transportation cost, carbon trading cost, and transportation time, a multiobjective 0–1 mathematical model is

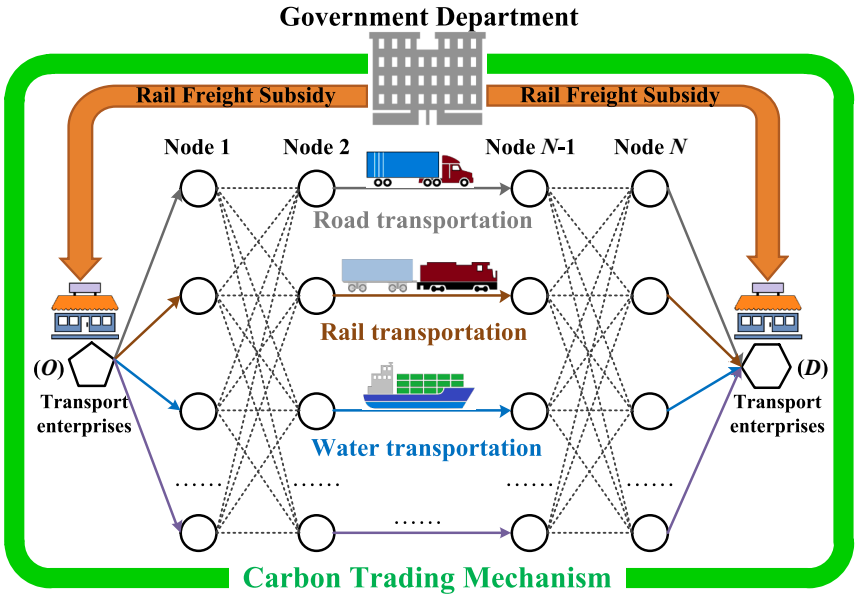


Fig. 2. Transportation network under the influence of a rail freight subsidy and carbon trading mechanism.

Table 2
Notations.

Notation	Description
Symbols	
i, j	Different nodes in the freight transportation network
k, l	Different transport modes, where 1 indicates road transportation, 2 indicates rail transportation, and 3 indicates water transportation
Sets	
M	Set of transportation network intermediate nodes
N_i	Set of all nodes connected to node i in the transportation network
S	Set of transportation network node arcs,$(i,j) \in S$
K	Set of transport modes,$k,\ l \in K$
K_i	Set of transport modes connected to node i
K_{ij}	Set of transport modes for transport network arc $(i.j)$
Parameters	
Q	Total freight volume
c_{ij}^k	Transport cost per unit turnover when adopting k mode of transport between nodes i and j
D_{ij}^k	Distance between nodes i and j when adopting k mode of transport
c_s	Unit freight subsidy for rail transportation
c_i^{kl}	Unit transshipment cost for converting k mode of transport to number l mode of transport at node i
e_{ij}^k	Carbon emissions per unit turnover when adopting k mode of transport between nodes i and j
e_i^{kl}	Unit transshipment carbon emissions of converting from k mode of transport to l mode of transport at node i
P_{ij}^k	Path's capacity between nodes i and j when adopting k mode of transport
R_i^{kl}	Transshipment capacity for converting from k mode of transport to l mode of transport at node i
E_q	Carbon quota for a transport enterprise
C_p	Carbon trading price
t_{ij}^k	Transportation time between nodes i and j with k mode of transport
t_i^{kl}	Transshipment time to convert from k mode of transport to l mode of transport at node i
Variables	
x_{ij}^k	For k mode of transport between nodes i and j, the value is 1; otherwise, the value is 0.
y_i^{kl}	For converting from k mode of transport to l mode of transport at node i, the value is 1; otherwise, the value is 0.

constructed to simultaneously consider the above three objectives.

(1) Minimize total transportation cost: min C

The total transportation cost C mainly includes the transportation cost of the different modes, the rail freight subsidy and transshipment cost. The transportation cost depends on the total freight volume Q, transport cost per unit turnover c_{ij}^k and transportation distance D_{ij}^k. The rail freight subsidy depends on the unit freight subsidy rate c_s when adopting rail transportation. In the current specification, freight transshipment are those operations that involve the transshipment between different modes of transport at an intermediate node. The transshipment cost is determined by total freight volume Q and the unit cost of the transshipment modes c_i^{kl}. The total cost of transportation can thus be expressed as

$$\min C = \sum_{(i,j)\in S}\sum_{k\in K_{ij}} Q c_{ij}^k D_{ij}^k x_{ij}^k - c_s \sum_{(i,j)\in S} Q c_{ij}^{Rail} D_{ij}^{Rail} x_{ij}^{Rail} + \sum_{i\in M}\sum_{k\in K_i}\sum_{l\in K_i} Q c_i^{kl} y_i^{kl} \tag{1}$$

(2) Minimize carbon trading cost: min E

The carbon trading cost E is the difference between the actual carbon emissions produced by freight transportation and the carbon quotas for transport enterprise E_q multiplied by the carbon trading price C_p. Carbon emissions are mainly the CO_2 emissions resulting from energy consumption. Intermodal transportation carbon emissions include transportation carbon emissions and transshipment carbon emissions, where transportation carbon emissions are determined by total freight volume Q, carbon emissions per unit turnover of the transport modes e_{ij}^k and distance D_{ij}^k, and transshipment carbon emissions are determined by total freight volume Q and unit carbon emissions of the transshipment modes e_i^{kl}. The carbon trading cost can thus be expressed as

$$\min E = \left(\sum_{(i,j)\in S}\sum_{k\in K_{ij}} Q e_{ij}^k D_{ij}^k x_{ij}^k + \sum_{i\in M}\sum_{k\in K_i}\sum_{l\in K_i} Q e_i^{kl} y_i^{kl} - E_q\right) C_p \tag{2}$$

(3) Minimize total transportation time: min T

Transportation time determines transportation efficiency and thus influences shippers' choices. Total transportation time includes transportation time and transshipment time. Transportation time depends on the distance D_{ij}^{k} and speed of the transport modes. Transshipment time is the sum of the transshipment times t_i^{kl}. Thus, total transportation time can be expressed as

$$\min T = \sum_{(i,j)\in S}\sum_{k\in K_{ij}} t_{ij}^{k} x_{ij}^{k} + \sum_{i\in M}\sum_{k\in K_i}\sum_{l\in K_i} t_i^{kl} y_i^{kl} \tag{3}$$

3.3.2. Constraints

$$\sum_{m\in N_i}\sum_{k\in K_{ij}} x_{mi}^{k} - \sum_{j\in N_i}\sum_{l\in K_{ij}} x_{ij}^{l} = \begin{cases} 1, i = d \\ 0, \forall i \in M \\ -1, i = o \end{cases} \tag{4}$$

$$\sum_{k\in K_{ij}} x_{ij}^{k} \leqslant 1, \forall (i,j) \in S \tag{5}$$

$$\sum_{k\in K_i}\sum_{l\in K_i} y_i^{kl} \leqslant 1, \forall i \in M \tag{6}$$

$$0 \leqslant c_s \leqslant \max c_s \leqslant 1 \tag{7}$$

$$\sum_{j\in N_i} x_{ij}^{l} = \sum_{k\in K_i} y_i^{kl}, \ \forall i \in M, \forall l \in K_i \tag{8}$$

$$\sum_{n\in N_i} x_{ni}^{k} = \sum_{l\in K_i} y_i^{kl}, \ \forall i \in M, \forall k \in K_i \tag{9}$$

$$x_{ij}^{k} \cdot Q \leqslant P_{ij}^{k}, \forall (i,j) \in S, \forall k \in K_{ij} \tag{10}$$

$$y_i^{kl} \cdot Q \leqslant R_i^{kl}, \forall i \in M, \forall k, l \in K_i \tag{11}$$

$$y_i^{kl} = 0, \forall i \in \{o,d\}, \forall k, l \in K_i \tag{12}$$

$$x_{ij}^{k} \in \{0,1\}, \forall (i,j) \in S, \forall k \in K_{ij} \tag{13}$$

$$y_i^{kl} \in \{0,1\}, \forall i \in M, \forall k, l \in K_i \tag{14}$$

Constraint (4) ensures the balance of freight flow at each node. Constraint (5) ensures that at most one mode of transport can be used between two adjacent nodes. Constraint (6) ensures that at most one transshipment operation is performed at the same node. Constraint (7) indicates that the value of rail freight subsidy rate should be within a reasonable range. Constraints (8) and (9) ensure the freight flow is indivisible during transportation. Constraint (10) ensures that the freight volume between node i and j should be within the capacity of the selected k mode of transport. Constraint (11) ensures that the freight transshipment volume of each node should not exceed the corresponding transshipment capacity limitation. Constraint (12) indicates that no transshipment operation occurs at the origin and destination nodes. Constraints (13) and (14) are decision variables that take the value of 0 or 1.

4. Solution algorithm

Multiobjective optimization models for intermodal freight transportation networks belong to the NP-hard problem (Ayar and Yaman, 2011), and it is difficult to find an optimal solution that optimizes all objective functions at the same time. To mitigate the complexity of solving multiobjective problems, the linear weighted sum method is usually used to set the weight coefficient of each objective function, and thus the multiobjective model is converted into a single objective model (Wang et al., 2018). However, the linear weighted sum method requires setting and quantifying the importance of each objective function (Sun, 2020), and the importance of different objectives is often not supported by relevant information, so objectively quantifying them is difficult. Moreover, the solutions obtained by this method cannot illustrate the effect of each objective function on the results. Therefore, the NSGA-II algorithm, which involves an elite strategy and a fast non-dominated ranking operation (Deb et al., 2002; Yin et al., 2021b), is chosen to solve the multiobjective problem and consider the hypersurface determined by the set of non-dominated solutions as the Pareto optimal frontier. In the application process, first the numerical results of all the solutions are calculated and non-dominated sorting is performed to eliminate the solutions that are already dominated. Then the first-level non-dominated solution set is obtained combined with crowding distance calculation. Finally, each solution in this set is analyzed and the evaluation function is used to obtain the optimal solution.

4.1. Evaluation methodology

In the obtained set of non-dominated solutions, there may be numerous Pareto frontier solutions, but none of these is better in all of the objectives. Most studies in the literature select the optimal solution by setting the evaluation function. This investigation adopts the ideal point method, the basic principle of which is constructing an evaluation function and selecting an ideal point that has the shortest sum of distances from each optimal solution, and the quality of the solution is measured by the proximity between the evaluation object and the ideal target. The evaluation function of the three-objective optimization problem is shown in Equation (15).

$$y = \sqrt{(f_1 - f_1^*)^2 + (f_2 - f_2^*)^2 + (f_3 - f_3^*)^2} \tag{15}$$

In Equation (15), f_1^*, f_2^* and f_3^* represent the optimal values of the individual objectives in the optimization model. They are the minimum transportation cost C^*, minimum carbon trading cost E^* and minimum transportation time T^*. Each part of this evaluation function represents the distance between the actual and optimal values of the individual objectives in each solution. The optimal solution is the solution where the actual values of the three objectives are closest to their ideal values, and thus the solution with the smallest evaluation index y is the optimal solution. In general, the Pareto solution is obtained as the optimal solution set, and obtaining the optimal solution intuitively is difficult. Therefore, the ideal point method is more feasible and intuitive to obtain the optimal solution.

4.2. Model solution

NSGA-II algorithm has the same process of selection, crossover and mutation operations compared with GA algorithm, the main difference is that after initializing the population and generating the first generation of the population, NSGA-II algorithm adopts an elitist strategy by selecting the better filial and parent generations to be recombined into a new population based on the non-dominance relationship between individuals and the crowding degree. The process for obtaining the solution is shown in Fig. 3. Specifically, it is as follows:

Step 1: Collect relevant data and write the objective function to calculate the optimal solution C^*, E^* and T^* under each goal.
Step 2: Combined with the non-dominated sorting algorithm to stratify the population size and obtain the first level of the non-dominated layer.

(1) Initialize the population.
(2) Perform selection, crossover and mutation operations and produce filial populations.
(3) Perform iterative operations for $Gen = 1$.
(4) Adopt the elite retention strategy to merge the filial and parent generations.
(5) Perform fast non-dominated sorting operations.
(6) Calculate the crowding distance.

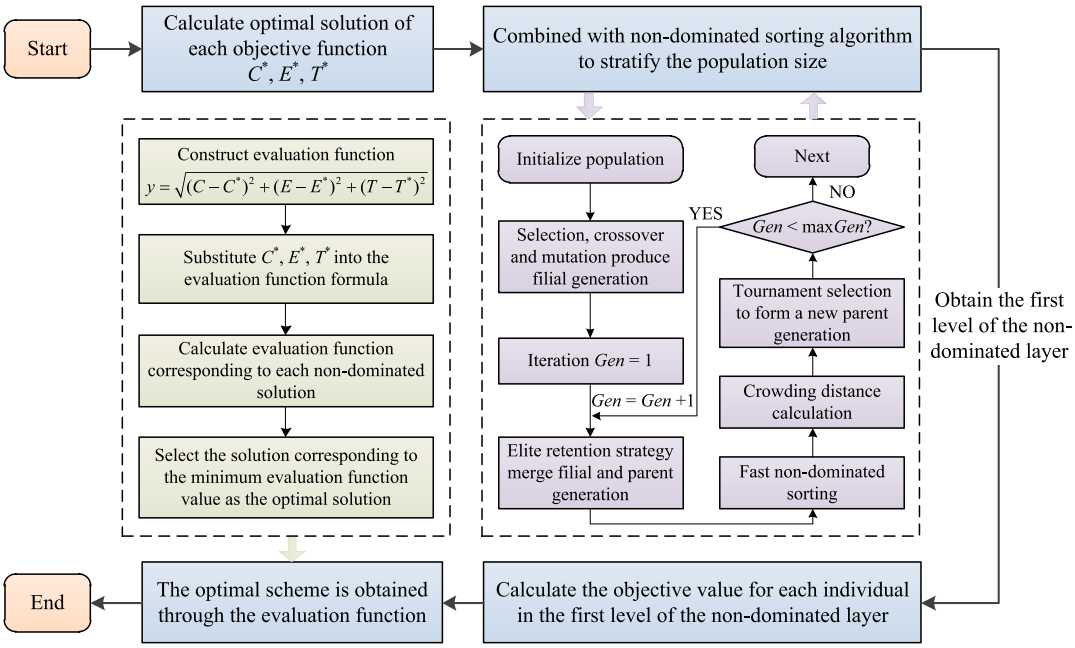


Fig. 3. Solution algorithm process chart.

(7) Perform tournament selection to form a new parent population.
(8) If *Gen* reaches max*Gen*, non-dominated individuals have been found; otherwise, let $Gen = Gen + 1$ and go to Step 2, Item (4).

Step 3: Calculate the objective value for each individual in the first level of the non-dominated layer.

Step 4: The optimal scheme is obtained through the evaluation function of the ideal point method.

(1) Construct the evaluation function $y = \sqrt{(C - C^*)^2 + (E - E^*)^2 + (T - T^*)^2}$.
(2) Substitute C^*, E^* and T^* into the evaluation function formula.
(3) Calculate the evaluation function corresponding to each non-dominated solution.
(4) Select the solution corresponding to the minimum evaluation function value as the optimal solution.

5. Numerical experiments

To conduct numerical experiments, Subsection 5.1 sets basic data and parameters in the model are set (Section 5.1). Subsection 5.2 applies a small-scale numerical example to analyze the efficiency and effectiveness of the NSGA-II algorithm (Section 5.2). Finally, a real-world case study in China is conducted to verify the application and practicality of the model (Section 5.3).

5.1. Data and parameter settings

5.1.1. Freight transportation data

According to the literature (Feng and Sun, 2022; Yin et al., 2023; Zhang et al., 2021c), data for the cost and speed of different transport modes and the time and cost of different transshipment modes are as listed in Table 3.

The rail freight subsidy is calculated with reference to the Chinese Ministry of Finance's limit on the CRE freight subsidy (not more than 30 % of the freight rate in 2020), and this study initially sets the rail freight subsidy at 5 %.

5.1.2. Carbon emissions data

The energy intensity and CO_2 emissions coefficient of the transport modes can be used to calculate the carbon emissions per unit turnover of freight transport. Because of the high proportion (about 73.8 % in 2022) of electrified railways in China, rail transportation is powered by electricity. As there is no air pollution associated with electric locomotives, this study considers the carbon emissions of the upstream power plants. Freight transportation by road is mainly by diesel trucks. According to the data in studies such as Tao et al. (2018), the energy intensity of electric locomotives is 0.122 kWh/TEU-km, and the energy intensity of diesel trucks is 0.168 L/TEU-km, which have a CO_2 emission factor of 3.758 kg/L. The electricity CO_2 emission factor is 0.895 kg/kWh, which is calculated from the average carbon emission factor for each regional power grid in China in 2019. The CO_2 emissions per unit turnover for water transportation is 0.26 kg/TEU-km according to Liu et al. (2018).

Carbon emissions from transshipment operations are closely related to transshipment conditions and equipment. Rail-mounted gantry cranes (RMGs) are mainly used for road to rail transshipment operations, while quay cranes are mainly used for road to waterway and waterway to rail transshipment operations. The unit energy intensity of an electric RMG is 3.150 kWh/TEU (Tao and Wu, 2021), and the unit energy intensity of a quay crane is 3 kWh/TEU (Yang et al., 2018). Based on these data, the unit transportation and transshipment carbon emissions are shown in Table 4.

5.1.3. Carbon trading data

Currently, China's carbon trading market lacks unified carbon quota allocation standards across different provinces and lacks objective data to support the determination of carbon quotas for intermodal transport enterprises. As rail transportation has significant advantages, such as low energy consumption and low pollution, it is the prime target for optimizing the transportation structure. Therefore, the carbon emissions from transporting a 20-ft container by rail are used as the basis for carbon quotas allocated to transport enterprises in this study. The carbon emissions per unit turnover of rail transportation is 0.109 kg/TEU-km. Thus, the carbon quota data can be determined by calculating the total freight volume, carbon emissions per unit of freight turnover and transport distances. This study adopts the free allocation method. If an enterprise's actual carbon emissions exceed this quota, it must purchase carbon emission rights through the carbon trading market. According to ICF International Consulting's "2021 China Carbon Price Survey

Table 3
Freight transportation data.

Transport mode	Unit transportation cost (¥/TEU-km)	Transport speed (km/h)
Rail	1.5	60
Road	6	70
Water	1.784	28
Transshipment mode	**Unit transshipment time (min/TEU)**	**Unit transshipment cost (¥/TEU)**
Rail–Road	4	5
Rail–Water	8	7
Road–Water	6	10

Table 4
Carbon emissions data.

Transport mode	Unit transport carbon emissions (kg/TEU-km)	Transshipment mode	Unit transshipment carbon emissions (kg/TEU)
Rail	0.109	Rail–Road	2.819
Road	0.631	Rail–Water	2.685
Water	0.260	Road–Water	2.685

Report", the average price in the national carbon trading market is expected to be 49 ¥/ton in 2022, rising to 87 ¥/ton in 2025 and 139 ¥/ton by 2030. The price in the 20 – 80th percentile range will increase from 30 ¥/ton – 50 ¥/ton in 2022 to 50 ¥/ton – 200 ¥/ton in 2030. Therefore, considering these predictions, this study sets the carbon trading price C_p as 50 ¥/ton (0.05 ¥/kg).

5.2. Numerical experiment on a small-scale network

5.2.1. Example description

A computational test is performed to analyze the efficiency and effectiveness of the NSGA-II algorithm. Fig. 4 shows an intermodal transportation network with 14 nodes, several randomly generated transport paths from (A) to (N) and three modes of transport: road, rail and inland waterway. The transportation distances between the nodes are shown in Table 5. Assuming that a batch of goods is loaded in 20-foot international standard containers, the freight volume is 50 TEU (twenty-foot equivalent unit), and the carbon quota for this transport operation is 4,000 kg. A transport enterprise needs to transport this batch of goods from origin node (A) to destination node (N).

5.2.2. Results of numerical example

To solve this numerical example, the population size is set to 50, the number of iterations is set to 200, the crossover rate is set to 0.9, and the variation rate is set to 0.1. The Pareto optimal solution set and the three objective values corresponding to each solution are obtained by the NSGA-II algorithm. The optimal values of the model's individual objective functions, transportation cost C, carbon trading cost E and transportation time T, are calculated as C^*, E^* and T^*, respectively, and used as the basis of the evaluation function. The evaluation function values are calculated with the ideal point method. The four indicator values of the seven specific solutions in this Pareto solution set are shown in Table 6.

As shown in Table 6, the transportation scheme with the smallest value for the evaluation function is Scheme # 1, which is regarded as the optimal transportation scheme. Its transportation modes and paths are shown in Fig. 5.

This optimal transportation scheme mainly uses rail and water transportation. The total carbon emissions are about 12,718.25 kg, and the carbon emissions per unit turnover are about 0.227 kg/TEU-km, which is almost 64.03 % less than the 0.631 kg/TEU-km unit carbon emissions of road transportation. The carbon emissions reduction is remarkable. With the joint effect of the rail freight subsidy and carbon trading mechanism, the transportation cost per unit turnover of this scheme is only 1.707 ¥/TEU-km compared with 6

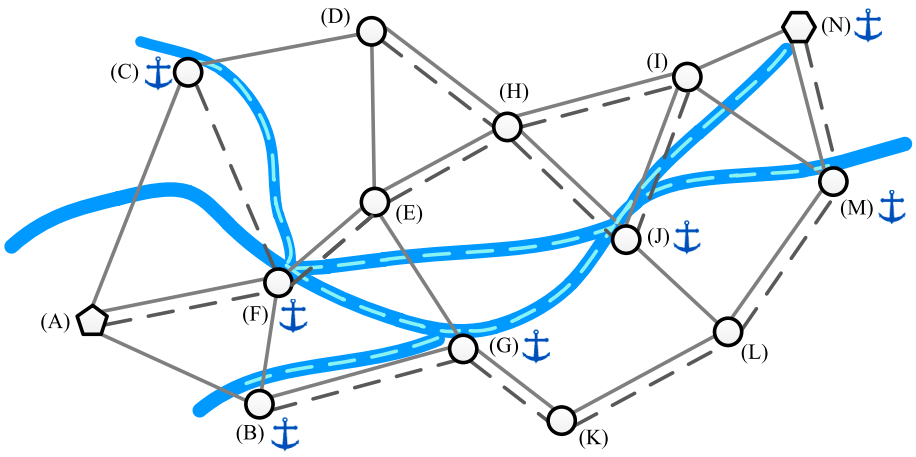


Fig. 4. Schematic diagram of an intermodal transportation network.

Table 5
Road/Rail/Water transportation distance from node *i* to *j* (Unit: km).

Segment	Distance	Segment	Distance	Segment	Distance
(A)~(B)	242/–/–	(E)~(G)	221/–/–	(I)~(M)	227/–/–
(A)~(C)	359/–/–	(E)~(H)	216/233/–	(I)~(N)	184/–/–
(A)~(F)	244/260/–	(F)~(E)	159/172/–	(J)~(I)	223/220/–
(B)~(F)	145/–/–	(F)~(G)	–/–/275	(J)~(L)	166/–/–
(B)~(G)	275/285/318	(F)~(J)	–/–/489	(J)~(M)	–/–/331
(C)~(D)	241/–/–	(G)~(J)	–/–/262	(J)~(N)	–/–/370
(C)~(F)	–/304/339	(G)~(K)	147/150/–	(K)~(L)	244/258/–
(D)~(E)	217/–/–	(H)~(I)	257/278/–	(L)~(M)	236/246/–
(D)~(H)	221/214/–	(H)~(J)	218/211/–	(M)~(N)	204/199/–

Note: "–" indicates no transport mode between the two nodes.

Table 6
Transportation scheme target and evaluation function value.

Number	Transportation cost (¥)	Carbon trading cost (¥)	Transportation time (h)	Evaluation function
1	95,497.8	435.9125	41.67857	152.3192
2	122,638.8	354.2750	21.67857	27,141.0900
3	318,000.0	1,472.1500	15.14286	222,505.4000
4	263,575.0	1,165.1380	19.32381	168,079.5000
5	179,931.3	683.9375	20.71667	84,434.4400
6	95,769.0	285.9225	34.48095	271.8886
7	228,130.0	961.1850	19.91905	132,633.9000

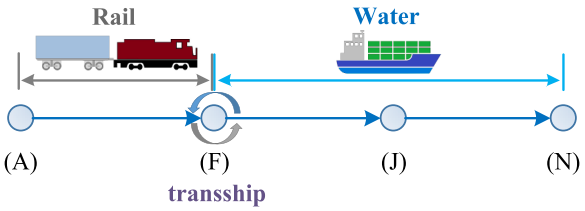


Fig. 5. Optimal transportation scheme of numerical example.

¥/TEU-km for all-road transportation.

5.2.3. Algorithm performance verification

To verify the performance of the NSGA-II algorithm, the GA algorithm designed with the ideal point method as the fitness function is compared with the NSGA-II algorithm to solve this multiobjective model. The average solving time of the NSGA-II algorithm and GA algorithm is compared on a small-scale transportation network. Besides, the values of the objective function under the operation of the NSGA-II algorithm and GA algorithm are observed after running these two algorithms 50 times each. The results of the algorithms comparison are shown in Table 7. In terms of the solving time of these two algorithms, with this small-scale numerical example, both the NSGA-II algorithm and the GA algorithm can obtain the results in a short period of time, and the solving time of the NSGA-II algorithm is shorter than that of the GA algorithm by about 1.9 s. Moreover, the NSGA-II algorithm is more advantageous in terms of the results of the objective function, and the average values of the objective functions of transportation cost, carbon trading cost, and transportation time are 9.2 %, 3.8 %, and 5.9 % less than those of the GA algorithm, respectively, which indicates that NSGA-II algorithm has better performance.

Fig. 6 represents the convergence process of these two algorithms in solving the different objective functions, after the algorithms run up to a certain number of iterations, they stabilize and remain near the optimal solution. As shown in Fig. 6, the GA algorithm reaches convergence around 120 generations, and the NSGA-II algorithm converges around 80 generations. Thus, the NSGA-II algorithm converges faster, and at the same time, the optimal values of the different objective functions obtained by NSGA-II algorithm

Table 7
Comparison of algorithm results.

Algorithm	Average solving time (s)	Average of objection function value		
		Transportation cost (¥)	Carbon trading cost (¥)	Transportation time (h)
NSGA-II	8.9	202,143.8	995.4	31.7
GA	10.8	237,299.2	1,034.5	33.7

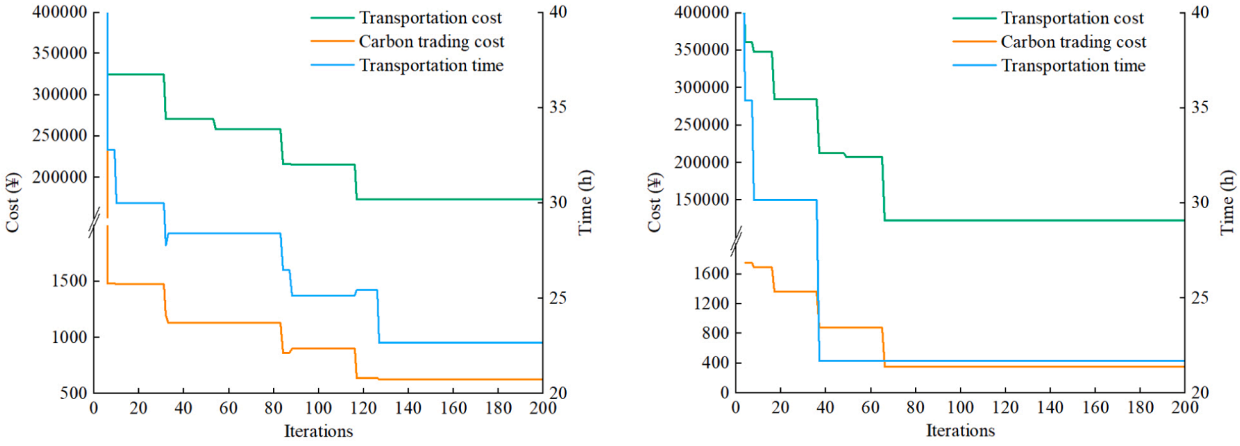


Fig. 6. Algorithm comparison of convergence process.

convergence are smaller than these obtained by the GA algorithm, which indicates that the NSGA-II algorithm is more efficient. Overall, the NSGA-II algorithm designed in this study converges faster, obtains better objective function values, and has a shorter solving time than the GA algorithm, thus, the NSGA-II algorithm has better algorithmic performance.

5.3. Case study

5.3.1. Case description

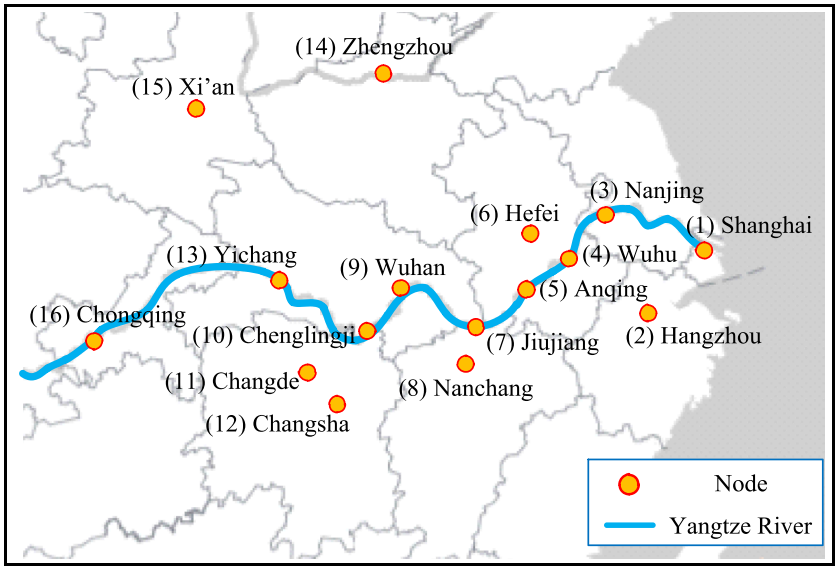
The Yangtze River is China's "golden" economic belt and occupies an important position in the process of China's economic development strategy. It accounts for approximately 20 % of China's land area, but its contribution to China's gross domestic product exceeds 40 % (Zhang et al., 2021a; Wu et al., 2022). At present, the transportation capacity of the Yangtze River Economic Belt region continues to increase, and the construction of road, rail and waterway transportation infrastructure is relatively complete, which has basically formed a well-integrated transportation system in China. However, through the analysis of the regional freight transportation structure of the Yangtze River Economic Belt, the current proportion of rail transportation is low, the proportion of road transportation is relatively high, and the transportation structure is still unreasonable. To investigate in depth the impact of rail freight subsidy and carbon trading mechanism on intermodal transportation network, the freight transportation from Shanghai to Chongqing is taken as the case and the neighboring cities along the Yangtze River Economic Belt are selected to construct an intermodal transportation network, including three modes of transport: road, rail and inland waterway. The schematic diagram of the intermodal transportation network of the Yangtze River Economic Belt is shown in Fig. 7.

To ensure the authenticity and practicality of case study results, relevant market data are collected, including the individual freight volume, transportation distance of different modes of transport, transportation time, carbon emission data, etc., from sources including China Railway Siyuan Survey and Design Group Company Limited and Yangtze River Coast Railway Group Company Limited. The Yangtze River Coast Railway Company is mainly responsible for coordinating intermodal transport business along the Yangtze River Economic Belt area. In this case, based on the company actual average daily statistics and a field market survey, the volume of freight transported in a single shipment from Shanghai to Chongqing is about 465 TEU. Referring to a survey and considering the company's actual operation and freight transportation business situation, the carbon quotas for this transport operation are about 90,000 kg. To focus on the key issues related to carbon emissions, the varying Yangtze River waterway channel congestion constraints are not explicitly modelled although they can be important practical considerations for specific inland waterways (Li et al., 2021).

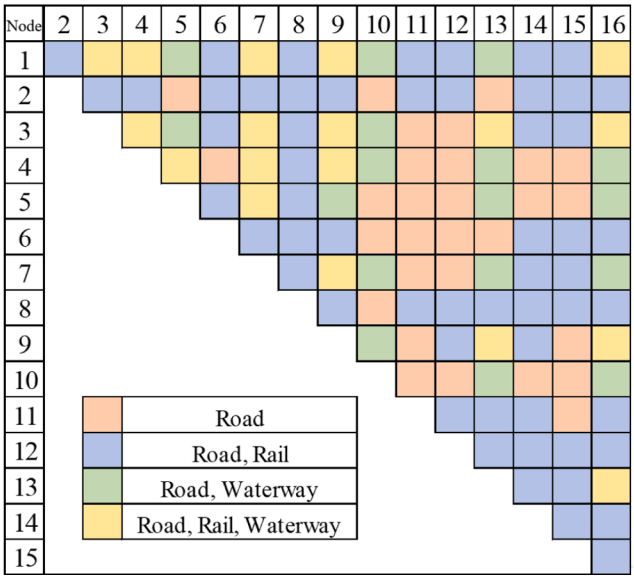
5.3.2. Results of case study

Based on the company's operation and field survey data, the NSGA-II algorithm is used to solve the mathematical model to obtain the Pareto optimal solution set for this case study, and the results are shown in Fig. 8. Each solution in the optimal solution set has different advantages in terms of different objective functions, and the optimal solution set can provide transport enterprises with different decision-making options and optimize the distribution of transport resources to achieve the trade-offs between the objectives of transportation cost, transportation carbon emissions and transportation time.

The ideal point method evaluation function is used to evaluate the solutions within the Pareto optimal solution set, and the optimal transportation scheme from Shanghai to Chongqing is obtained as shown in Fig. 9. The transportation cost is ¥1,498,457.85, the carbon trading cost is about ¥2,745.79, and the transportation time is about 107.18 h. The total carbon emissions are about 144,915.86 kg, while the carbon emissions per unit turnover are about 0.146 kg/TEU-km, which is almost 76.86 % less than the 0.631 kg/TEU-km unit carbon emissions of road transportation. The carbon emissions reduction is quite remarkable, which is also in line with the realistic needs of low-carbon orientation of transport enterprise.



(a) Transportation network nodes



(b) Transport mode

Fig. 7. Intermodal transportation network of the Yangtze River Economic Belt.

With the gradual improvement of the construction of the rail lines along the Yangtze River in China, the rail freight capacity along the riverine area has been fully released, with the application of rail-water intermodal transport organization becoming a more common form of transport organization in the areas along the Yangtze River Economic Belt in China (Yin et al., 2023). However, it is worth noting that rail-water intermodal transportation is effective for long-distance bulk cargo or container transportation. For time-sensitive freight such as fast moving consumer goods (FMCG), it is still necessary to cooperate with roads and other modes of transport with faster transport speeds to further optimize the transport organization.

5.3.3. Sensitivity analysis

To further analyze the variations in intermodal transportation network schemes under the joint influence of a rail freight subsidy and carbon trading mechanism, the NSGA-II algorithm and the ideal point method evaluation function are used to calculate and observe numerical changes in the transportation cost, carbon trading cost and transportation time of the optimal scheme under various scenarios.

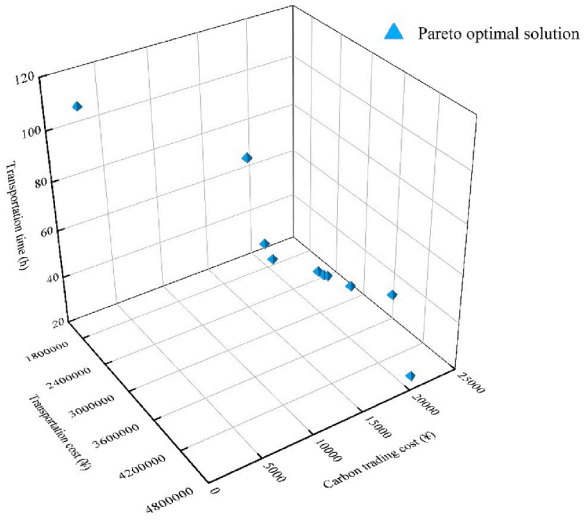


Fig. 8. Pareto optimal solution set of case study.

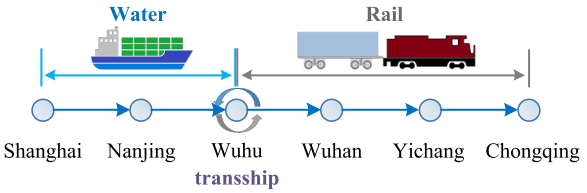


Fig. 9. Optimal transportation scheme of case study.

(1) Sensitivity analysis of the rail freight subsidy

With the carbon quota and carbon trading price unchanged, the relation between the optimal transportation scheme and rail freight subsidy is obtained by altering the rail freight subsidy rate from 5 % to 30 % (in increments of 5 %), as shown in Fig. 10.

As shown in Fig. 10, when the rail freight subsidy is increased from 5 % to 10 %, the optimal transportation scheme changes, and the transportation cost decreases to ¥1,430,014.50, the carbon trading cost decreases to ¥1,273.02 and the transportation time is shortened to 37.97 h. Further increases in the rail freight subsidy generally only decrease the transportation cost, while the optimal transportation scheme remains unchanged. Thus, the optimal rail freight subsidy rate is 10 % with the other parameters of the model unchanged, and the corresponding optimal transportation scheme is as shown in Fig. 11.

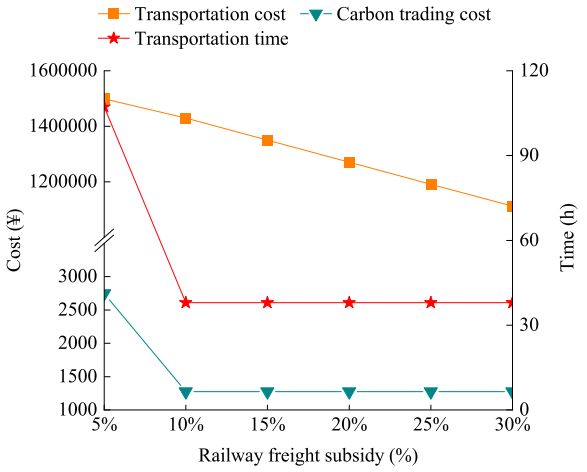


Fig. 10. Analysis of the rail freight subsidy.

As shown in Fig. 11, when the rail freight subsidy is 10 %, enterprises tend to choose direct rail transportation, and the freight transportation structure is further optimized. The total carbon emissions of this transportation scheme are 115,460.43 kg. Compared with the transportation scheme in Fig. 9, carbon emissions are reduced 20.33 %, carbon emissions per unit turnover decrease 25.34 %, and the cost per unit turnover decreases 7.36 %. Furthermore, the total transportation time was effectively reduced as a result of the adoption of the direct rail transportation, which avoided freight transshipment operations. Therefore, the introduction of a rail freight subsidy would significantly help optimize the freight transportation structure by reducing transportation costs and emissions.

(2) Sensitivity analysis of the carbon emissions quota

The optimal transportation scheme under different carbon quotas is calculated by increasing the carbon emissions quota from 50,000 kg to 90,000 kg (in increments of 10,000 kg), as shown in Fig. 12.

According to the results in Fig. 12 and in contrast with the previous experience, the optimal transportation scheme does not change significantly with continuous limitations in the carbon quota. In addition, from a theoretical perspective, the cost of carbon emissions quota as a constant term of the objective function, it does not affect the scheme decision of transport enterprise significantly. Although the carbon trading cost gradually increases as the carbon emissions quota decreases and enterprises must purchase excess carbon emission rights through the carbon trading market, compared with the transportation cost, the carbon trading cost remains low when the carbon trading price is maintained at 0.05 ¥/kg. Even though the carbon trading cost gradually increases, the increase is not sufficient to affect transportation scheme decisions.

(3) Sensitivity analysis of the carbon trading price

Carbon trading prices in China differ significantly between and within regions. From 2020 to 2021, the carbon price in Beijing was mostly above 0.08 ¥/kg. The carbon price in Shanghai fluctuated around 0.04 ¥/kg throughout the year, while the carbon price in Fujian fell below 0.01 ¥/kg at the end of 2020 and then gradually rose to nearly 0.02 ¥/kg in 2021. Therefore, a sensitivity analysis of the carbon trading price is necessary. The optimal transportation scheme under different carbon trading prices is calculated by increasing the carbon trading price from 0.05 ¥/kg to 0.55 ¥/kg (in increments of 0.05 ¥/kg), as shown in Fig. 13.

As shown, when the carbon trading price increases from 0.35 ¥/kg to 0.40 ¥/kg, the optimal transportation scheme changes, which is the same result as in Fig. 11. When the carbon trading price increases, enterprises spend more to purchase carbon emission rights because their carbon quotas remain unchanged. Thus, rail transportation, with its substantial low-carbon and efficiency advantages is fully utilized, which helps enterprises reduce carbon emissions during freight transportation process and reduce carbon trading cost.

6. Discussion and policy implications

The previous investigation of the collaborative effects of a rail freight subsidy and carbon trading mechanism on the transportation network shows that a freight subsidy for low-carbon transportation systems could help optimize the transportation structure and promote the sustainable development of intermodal transportation. Meanwhile, the carbon trading mechanism is an important policy tool to control and reduce emissions in the transportation industry. Policy instruments can facilitate the shift to low-emission modes of transport and promote a low-emission future for the industry (Yan et al., 2021). The combined effect of a freight subsidy and carbon trading mechanism can contribute to the construction of an efficient, low-carbon, and integrated transportation network. It is also found that, when the carbon trading price is low, the carbon trading mechanism has a limited effect on the enterprises' decisions regarding transport schemes, which contrasts with the previous findings and indicates that there is extensible space for improvement in the current carbon trading mechanism. To further optimize the intermodal transportation structure and realize the low-carbon and sustainable transformation of the transportation industry, the following policy implications are put forward and discussed.

The optimal transportation scheme obtained from the case study in this paper shows that under the collaborative effects of rail freight subsidy and carbon trading mechanism, the transportation scheme often adopts rail-water intermodal transportation (as shown in Fig. 9) or direct rail transportation (as shown in Fig. 11). Although the transportation cost and carbon emissions can be effectively controlled and reduced by using rail-water intermodal transportation, there is still a significant gap between its timeliness and that of road or direct rail transportation, which is a challenge for the freight transportation of FMCG and other goods with high timeliness requirements. In addition, when the proportion of rail freight subsidy rises to 10 % or the carbon trading price reaches 0.40 ¥/kg, the optimal transportation scheme will use direct rail transportation more often, which will effectively guarantee the timeliness of freight transportation. Therefore, policy insights are gained for different actors of transport services in this situation. For the final users of transport services, such as shippers and consignees, shippers can select the most suitable mode of transport according to the type,

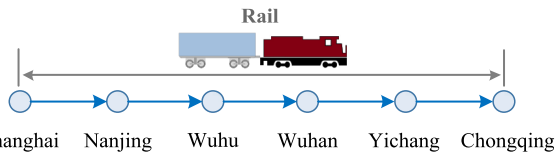


Fig. 11. Optimal transportation scheme with 10% rail freight subsidy.

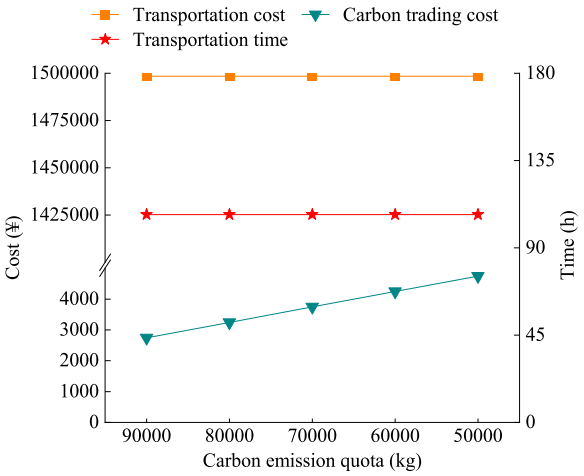


Fig. 12. Analysis of the carbon emissions quota.

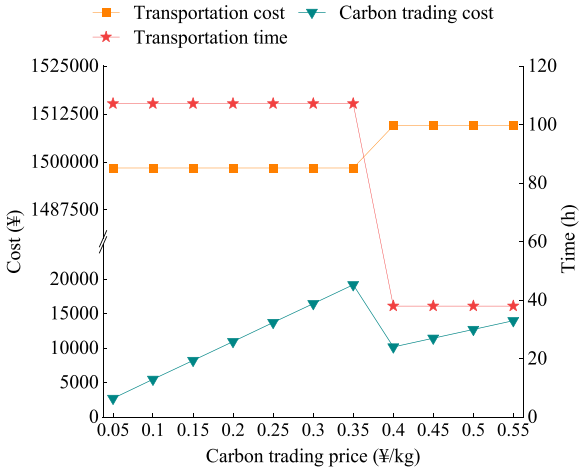


Fig. 13. Analysis of the carbon trading price.

quantity and time value of the cargo, fully consider the influence of freight subsidy policies and the carbon trading mechanism, and formulate the optimal transportation scheme with favorable economic and environmental benefits. For consignees, they can also pay attention to the dynamics of the relevant freight subsidy and carbon trading policies in a timely manner, and further adjust their procurement decisions in advance and reduce costs. For the intermediaries of transportation services, such as freight forwarders, they should optimize the utilization of vehicles for the freight transportation, such as the use of hydrogen-powered wagons, electric locomotives, etc., to further control carbon emissions in the transportation chain.

Based on case study and sensitivity analysis results, a rail freight subsidy can effectively improve the market competitiveness of rail transportation, shift freight flows from roads to rails, and help realize the rationalization and low-carbon transformation of the transportation structure. In this study, the optimal rail freight subsidy rate is 10 %. If the rail freight subsidy is increased further, total transportation cost is further reduced, but the financial pressure on governments gradually increases. Besides, the current freight subsidy policy often lacks sustainability, and the share of intermodal transportation may decline once the freight subsidy is removed. Therefore, to promote the sustainable development of low-carbon intermodal transportation, it is vital to rationalize the design of freight subsidy policies for low-carbon modes of transport, and the government can adjust the structure of the freight market through price leverage and the rail freight subsidy mechanism. In the future, a waterway freight subsidy to address congestion and safety on inland waterways could be implemented to shift freight flows from roads to waterways, further enhancing the use of low-carbon transportation modes and reducing the carbon emissions of the integrated transportation system.

Meanwhile, carbon trading price also has an important impact on intermodal transportation network design. However, the intra- and inter-regional differences in carbon trading prices in China have high variability and volatility. Taking Shenzhen, China's first carbon trading market, as an example, its historical high carbon price reached 130.90 ¥/ton, while the lowest price was only 15.67 ¥/ton. The optimal carbon trading price in this study is about 400 ¥/ton, which significantly differs from the current average domestic carbon trading price, but is in line with the current carbon trading price level in the European region (Zhang et al., 2022; Sun et al.,

2023). Therefore, price leverage can be used to guide enterprises to reduce carbon emissions and optimize resource allocation. At higher carbon prices, an enterprise must follow the strategy that annual emissions do not exceed its carbon quota (Huang et al., 2023b), however, both high and low carbon trading prices have drawbacks. A low carbon trading price reduces enterprises' motivation to reduce their emissions voluntarily, whereas a high carbon trading price puts an excessive economic burden on some high-emission enterprises (Cheng et al., 2017; Babagolzadeh et al., 2020). Carbon pricing is the key factor in achieving the emissions reduction target (Tang et al., 2020), and government departments should consider market supply and demand, general trends of industry development and other factors in guiding the carbon trading market to develop a reasonable pricing mechanism and guaranteeing its stable operation.

The previous experiments show that carbon quotas have a limited effect on intermodal transportation schemes when the carbon trading price is relatively low, so enterprises are not motivated to reduce carbon emissions. Therefore, in addition to rationalizing the carbon trading price using market mechanisms, a freight subsidy mechanism for low-carbon transportation modes can be introduced at an appropriate time. The freight subsidy mechanism can promote the low-carbon and sustainable transformation of the transportation structure, and this is consistent with the intention that the transportation industry participate in the carbon trading mechanism. Thus, optimizing the intermodal transportation network through the coordination of a freight subsidy and carbon trading mechanism is also an effective first step toward including the transportation industry in the carbon trading system.

In addition, government departments should improve the carbon quota policy for the transportation industry. The allocation of carbon quotas is crucial for reducing carbon emissions (Qiu et al., 2017). Currently, some Chinese provinces and regions include the transportation industry in the carbon trading mechanism. However, the regional implementation of carbon quotas, carbon prices and the trading mechanism is not entirely consistent. Meanwhile, carbon emissions in the transportation industry are characterized by irregular boundaries and multiple and mixed sources. The current carbon trading mechanism mostly targets fixed emission sources in the transportation industry, while a unified system for mobile emission sources, such as road vehicles and locomotives, is lacking. In terms of the manner of carbon allowance allocation, China government uses a mixed approach to allocating carbon quotas, with free allocation as the main method and paid allocation as a supplement. In the long-term, free carbon quotas are not conducive to the formation of a market pricing mechanism, and thus they reduce the operational efficiency of the carbon trading market. Therefore, government departments should improve the carbon emissions accounting system by investigating carbon quotas in the transportation industry and establishing a scientific and practical method of allocating carbon quotas. Further, the emissions reduction potential of the transportation industry should be analyzed to avoid over-allocation while considering the ability of different types of transportation enterprises to afford additional carbon credits, and a carbon quota allocation method in line with the future development demands of the transportation industry should be promulgated.

Finally, for the transferability and applicability of the findings, it is worthwhile to promote the joint mechanism of rail freight subsidy and carbon trading mechanism in different countries and regions beyond China. The carbon trading market in China is still at the early stage of its development. As illustrated in the case study, the current carbon trading price seems insufficient to influence the market structure of freight transportation. Promoting the collaboration between rail freight subsidy and carbon trading mechanism can make up for the shortcomings of the carbon trading mechanism to a certain extent, and more effectively promote the low-carbon transformation and sustainable development of freight transportation. As for other regions such as the European Union, where the carbon trading mechanism is more well developed and implemented, perfecting the subsidy mechanism for low-carbon transport modes can not only help the country's transportation industry to achieve the "net-zero carbon emission" plan, but also further optimize the freight transportation structure of the logistics market, so as to reduce the cost of international logistics, and enhance the international competitiveness of the trade in transportation services.

7. Concluding remarks

This study constructs a transportation network optimization model with coordinated rail freight subsidies and a carbon trading mechanism. The results show that the combination of the NSGA-II algorithm and the ideal point method is of great practical value in solving multiobjective optimization problems. This method avoids the subjectivity of the multiobjective linear weighted sum method and can also solve difficult decision problems owing to the complexity of the Pareto solution obtained from the multiobjective solution.

Additionally, the collaborative effects of a freight subsidy and carbon trading mechanism on the optimization of the transportation network is relatively remarkable. In the context of an unbalanced transportation structure, freight subsidies for low-carbon transportation modes help relieve enterprises' operating pressure and help promote the development of low-carbon and energy-efficient transportation modes, actively shifting freight flows from roads to railways or waterways and rationalizing the transportation structure. Meanwhile, the scientific and reasonable formulation of a carbon trading price also has an obvious role in promoting the optimization of the freight transportation structure, and appropriate carbon pricing will force enterprises to adjust the transportation structure and upgrade their facilities and equipment to reduce their carbon emission cost. However, in contrast with previous studies, this study shows that the effect of carbon quotas on the intermodal transportation scheme is relatively limited, because the carbon trading price in China remains at a low level and enterprises' carbon quotas are mostly free, which weakens the cost constraint effect and undermines its ability to influence the transportation structure decisions of transport enterprises.

The meaningful policy implications have been obtained for practice that can improve sustainability of the transportation industry. Different actors in the transport service should accelerate their adaptation to the trend of carbon emissions reduction in the industry with the guidance of relevant low-carbon policies. And government departments should improve the formulation of sustainable freight subsidy policies based on a full investigation of the current situation of the freight transportation market. In addition, the carbon trading price and carbon quota allocation mechanism should be effectively refined, considering the emission reduction potential of the

industry and the actual operating capacity of enterprises. The experience of promoting energy conservation and emission reduction in the transportation industry through the collaboration of a rail freight subsidy and carbon trading mechanism is worth expanding to different countries.

The work in this paper also has some limitations: the carbon trading mechanism for the transportation industry is still in an early stage of development; its coverage varies by region, and there is no unified carbon quota allocation and accounting system. Thus, the availability of carbon emissions data for transport enterprises is limited. Therefore, the carbon quotas allocation is determined by estimating the carbon emissions reduction potential of transport enterprises in this study, which inevitably leaves a gap between the research results and reality. In addition, a long-term freight subsidy policy would be a heavy financial burden for local governments, and this study does not consider the timely withdrawal of the rail freight subsidy.

Finally, further optimization directions are also identified. To further the continuous improvement of the carbon trading system, studies could examine more detailed empirical data on different transport enterprises, and super-heuristic algorithms could be designed to optimize the intermodal transportation network considering the uncertainty in the carbon trading market, which can also improve the efficiency of mathematical model solving and provide more decision supports for relevant policy makers. In view of the regional characteristics of China's carbon trading system, carbon quotas for mobile transportation devices and their impact on the transportation network could also be studied.

CRediT authorship contribution statement

Chuanzhong Yin: Writing – original draft, Validation, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Zi-Ang Zhang:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Xiaowen Fu:** Writing – review & editing, Validation, Supervision, Project administration, Formal analysis, Conceptualization. **Ying-En Ge:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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